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MobiScale Incubator

MobilityTech in CEE: Investment and Innovation landscape

Strategic Report - 2026

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Methodology note

Due to the profile of the report's audience, which includes individuals responsible for managing companies and institutions in the mobility sector, the analysis focuses on the key directions of sector development and their strategic implications. The starting point was a set of initial assumptions based on market knowledge and the authors' experience, which guided the subsequent analytical work.

The report is based on the analysis of secondary data and its validation through interviews with market experts. Quantitative data comes from industry reports as well as raw data analysis from Dealroom and Eurostat databases.

Hypotheses formulated during the analysis were tested through quantitative data analysis and cross checked with the perspectives of investors, founders, and industry representatives. A hypothesis was considered confirmed when data driven insights were consistent with expert opinions.

A limitation of the analysis is the non uniform classification of transactions across databases such as Mobility and Transportation. As a result, the analysis applies expert based standardization. Additionally, part of innovation in TSL takes place outside the VC channel including corporate budgets, integrators, and grant funded projects which are not fully captured in the data but are reflected in expert interviews.

In this report, MobilityTech is defined as technologies for transport, logistics, and mobility across both B2B and B2C models, while the CEE region is defined as Central and Eastern European countries that are members of the European Union: Poland, Czechia, Slovakia, Hungary, Romania, Bulgaria, Croatia, Slovenia, Lithuania, Latvia, and Estonia, in line with classifications used in Dealroom and Eurostat databases.

Executive Summary

CEE is one of the key operational hubs of European transport and logistics: the region accounts for 28% of EU territory and 22% of its population, with the TSL sector contributing on average 6.2% of GDP (Poland: 6.7%), compared to 4.6% in Germany and 4.2% in France.

However, this operational weight does not translate into proportional capture of technological value. Despite a significant number of funded companies, the region attracts only around 5% of total VC investment value, and in MobilityTech the CEE share remains heavily dependent on individual mega rounds.

In practice, this means that CEE operates as an executor of operations, but less often as an owner of products and technological infrastructure that monetize those operations.

Share of the value of venture capital investments made in Central and Eastern European countries in the total value of venture capital investments in the European Union, %
The data covers venture capital investments from 2019 to 2024.
Source: Dealroom, own calculations

Executive Summary

The diagnosis indicates that the main barrier is not a lack of capabilities, but structural constraints: low margins (2 to 5% net), market fragmentation, limited CAPEX, and low maturity of innovation processes on the side of both technology adopters and funds. In such an environment, the most rational investment path begins where returns are fastest and repeatability can be built: in the standardization of micro processes, automation, and AI deployment in back office and operations. Reducing implementation risk is critical, through standardized pilots, partnerships, and risk sharing mechanisms between operators and technology providers.

In the long term, the region needs not only greater capital supply, but also market infrastructure that removes scaling bottlenecks: access to pilot environments, data interoperability, integration capabilities, and enterprise deployment channels. Without such a system, the most promising projects will relocate as they scale, and the competitive gap with Western markets will continue to widen alongside increasing labor cost and regulatory pressures.

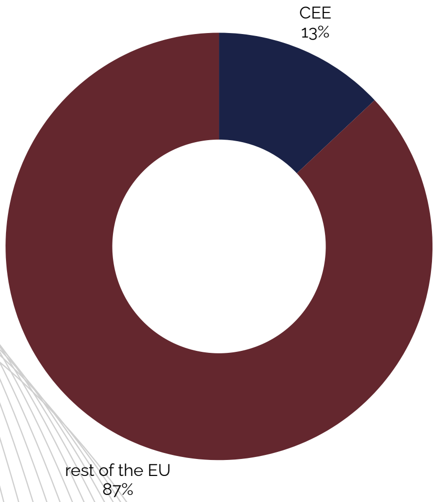
Report context

1.1 CEE as a logistics and transport hub in the European Union

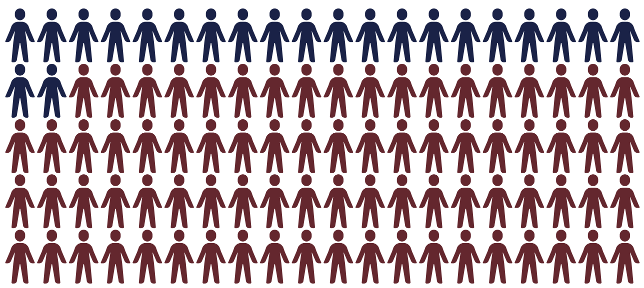
The Central and Eastern European region accounts for 28% of the European Union's territory and 22% of its population, yet generates only 13% of the EU's total GDP. This disparity highlights the region's economic potential, which despite its scale remains relatively underinvested.

At the same time, the region has developed a clear specialization in transport, freight, and logistics. According to Eurostat data, the sector accounts for an average of 6.2% of GDP across CEE countries. Poland reaches 6.7%, making it one of the regional leaders alongside Lithuania at 10.9% and Romania at 7.3%. Compared to the largest EU economies, the region's advantage is even more pronounced, with the TSL sector contributing 4.6% of GDP in Germany and 4.2% in France, against an EU average of 4.8%.

Share of the total gross domestic product of Central and Eastern European countries in the overall gross domestic product of the European Union, %
Source: 2025 (nama_10_gdp), own calculations



Share of the population of Central and Eastern European countries in the total population of the European Union, %
Source: Eurostat 2025 (demo_pjan), own calculations

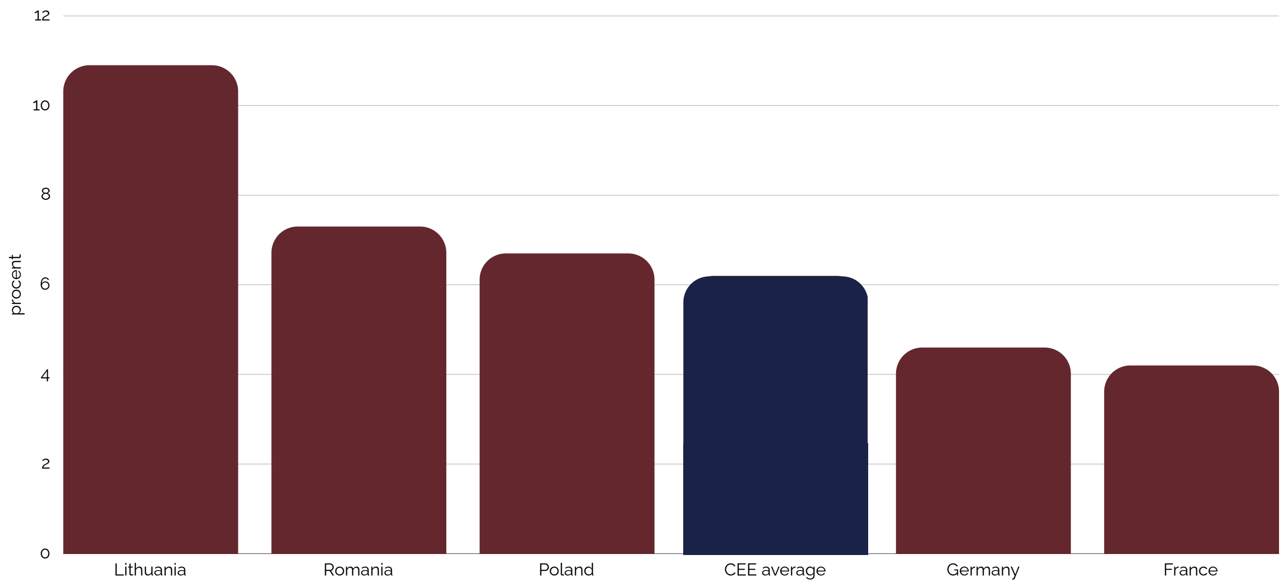


Report context

1.2 Overview of global MobilityTech trends and their implications for the sector in Central and Eastern Europe

In 2024, the mobility sector in Europe attracted USD 6 billion in funding, marking a 30% decline compared to the previous year. Despite this, it remains among the leading sectors and has consistently ranked in the top 5 VC industries on the continent for over a decade. This is driven by the fact that mobility sits at the intersection of megatrends such as decarbonization, electrification, and digitalization, which are shaping the transformation of the entire economy.

Share of the transport, freight forwarding, and logistics sector in gross domestic product in selected countries and regions, %. Comparison of Central and Eastern European countries with high specialization in the TSL sector and selected Western European economies with a more diversified economic structure.
Source: Eurostat 2023 (nama_10_a64), own calculations



Report context

While Western European countries such as Germany, France, and the Netherlands maintain stable and predictable levels of funding for projects in this sector, the CEE region is characterized by high volatility and relatively low significance compared to the overall EU market. Individual large rounds in countries such as Estonia, Croatia, or Slovakia can significantly increase the region's share of VC investment in mobility, reaching up to 9% of total EU mobility VC funding. Without them, however, this share drops to marginal levels, amounting to just over 5% of total VC investment in mobility across the EU, based on Dealroom data.

A clear example is Bolt, which raised EUR 628 million in its Series F round in 2022. The scale of this single round had a substantial impact on the total investment value in the CEE region.

This illustrates that capital in Europe is concentrated in mature Western and Northern hubs, which capture the vast majority of VC funding in the transportation sector, leaving CEE on the periphery.

Experts interviewed

ALEKSANDRA KASZUB
MARATHON INTERNATIONAL

ADAM ŻELAZKO
ROHLIG SUUS LOGISTICS

GRZEGORZ STAŃCZAK
EXPERT IN LOGISTICS SECTOR

MICHAŁ LASOCKI
EEC VENTURES

JAN RADZIKOWSKI
CAPITAL MARKET EXPERT

MICHAŁ WOLAŃSKI
SGH WARSAW SCHOOL OF ECONOMICS

PAVEL VRBA
EIT URBAN MOBILITY

PAWEŁ MICHALSKI
VC LEADERS

ŁUKASZ CHYLIŃSKI
GRUPA DBK

SEBASTIAN WRÓBEL
FREIGHT TECH

MATHIAS BOSSE
PREQUEL VENTURES

JAN RICKERS
MOBILITY FUND

NATALIA NOGACKA
BOOTSTRAPPIE

JOANNA PORATH
AC PORATH

MARCEL NAWALANY
BITCARGO

MACIEJ STOPCZYK
DEPOTRACER

DANIEL KIERDAL
CEE TECH

SEBASTIAN PRUCKMAIR
AEROMOND

KACPER NOWICKI
NOMAGIC

TOMASZ ZARZYCKI
EXPERT IN LOGISTICS SECTOR

MATTHIAS SCHANZE
RETHINK VENTURES

Hypotheses on key MobilityTech challenges in the CEE region

The numbering of hypotheses is for organizational purposes only and does not reflect their importance. The first digit indicates the area of analysis, while the second refers to a specific hypothesis. H1 relates to venture capital investments in CEE, H2 to market barriers, H3 to macro trends, H4 to directions of solution development, and H5 to capital availability and scaling potential.

H1.1 CEE region is underinvested in VC regardless of sector

Status: Confirmed | Confidence: High

The Central and Eastern European region, despite its growing position in the technology ecosystem, remains systematically undervalued in terms of capital allocation. According to the Dealroom report "Central and Eastern European Startups 2025" and analysis by The Recursive, CEE accounts for around 41% of all venture backed startups in the European Union, representing approximately 3,800 companies out of a total of 9,190. At the same time, the region attracts only 5% of the total VC investment value. This disparity persists regardless of sector.

Michał Wolański's comment:
„In Poland, we have very little capital, and even less that is willing to take risks. What we can do is focus on building small, smart businesses and, at a certain point, seek global funding. Low margins are a challenge, but an even bigger issue is the lack of capital surplus."

Paweł Michalski's comment:

„The phenomenon is driven by limited capital accumulation in the region, a relatively low level of financial surplus, and a scarce track record of large scale exits that could reinject both capital and experience into the ecosystem. There is a lack of individuals who, based on prior successes, could afford to take bolder risks, experiment with new models, and finance ambitious projects. As a result, growth dynamics fall short of the region's potential, and the development of the ecosystem is constrained by a shortage of capital."

Sebastian Wróbel's comment:

„The biggest constraint for mobility companies in CEE today is not a lack of technological capabilities, but the structure of financing. Historically, the region has not developed sufficient private capital accumulation or a strong base of angel investors willing to take risks in capital intensive projects. As a result, relocation becomes a natural path for some companies, particularly to the United States, where access to funding and market scale are greater. This is not an emotional decision, but an economic one."

Pavel Vrba's comment:

„As the Polish economy grows, its capacity to accumulate capital domestically also increases. A growing share of financial resources remains within the local economy, and these funds can, in the future, support a greater number of pilot projects, a higher willingness to take risks, and more ambitious innovation."

Jan Rickers' comment:

„Traditional capital, such as family offices that have built their wealth in conventional industries, often avoids venture capital, perceiving it as too risky. They typically invest in real estate or private equity funds, and are less likely to engage as early stage startup investors. As a result, a much more effective source of capital for young technology companies are founders who have previously achieved large exits and begin reinvesting in new ventures.

The emergence of a domestic startup ecosystem is often triggered by a first major exit. Former founders then start reinvesting in the local or national ecosystem and in subsequent startups. This creates an acceleration effect and enables the development of a new ecosystem.

I believe this is still, to some extent, missing in parts of Central Europe. In Germany, particularly in Berlin, large exits appeared relatively early. A notable example is the Samwer brothers, who built Rocket Internet along with their own incubator and investment fund. This served as a starting point for the ecosystem, which is one of the reasons why Germany is now such a strong market. A similar process could also take place in Central and Eastern Europe. Estonia may be an exception, as a major exit occurred there relatively early.

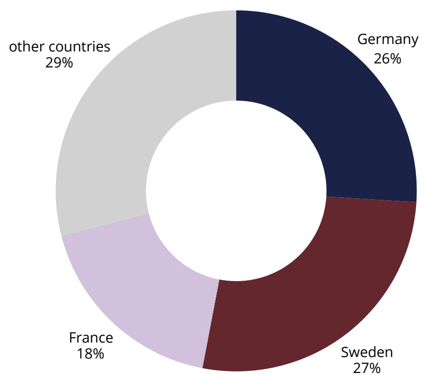
Access to funding is one of the factors that may encourage some founders to relocate to hubs such as London or Berlin. While they originate from Central and Eastern Europe, they may choose not to establish their startups there or decide to relocate after an initial funding round. As a result, founders may come from the CEE region, but the startup itself is classified as a Western European company, which likely also affects statistics on the number and value of startups in the region."

H1.2: VC investment in MobilityTech in CEE is significantly below the EU average

Status: Confirmed | Confidence: High

Based on Dealroom data for the period 2019 to 2024, CEE attracted EUR 4 billion out of EUR 44.68 billion in VC investment in transportation across the EU, representing less than 9%, despite the region generating 13% of EU GDP and 16% of transport value added. Excluding mega rounds, CEE attracted EUR 993 million out of EUR 19.27 billion across the EU, accounting for approximately 5.1% of the total.

Share of the value of venture capital investments in the MobilityTech sector by European Union countries, %
The data covers venture capital investments in the MobilityTech sector from 2019 to 2024, in the "transportation" segment, based on own analysis of Dealroom data.



The average level of investment in MobilityTech across the European Union is largely driven by a few of the largest and most mature ecosystems. Dealroom data for the period 2019 to 2024 shows that Germany accounts for approximately 26% of VC investment value in MobilityTech in the EU, Sweden for around 27%, and France for nearly 18%. This means that these three countries generate over 70% of all VC investment in the mobility sector in the EU.

Paweł Michalski's comment:

„One of the key challenges is the lack of truly global companies in the region, and consequently a lack of international experience, networks, and capabilities in building sales structures abroad. This translates not only into a skills gap, but also into a lack of global mindset and ambition needed to build companies attractive to VC investors and to effectively understand customer needs and sales dynamics in international markets."

Marcel Nawalany's comment:

„When it comes to angel investors, one of the challenges is that they rarely come from the transport, logistics, or broader MobilityTech sectors.

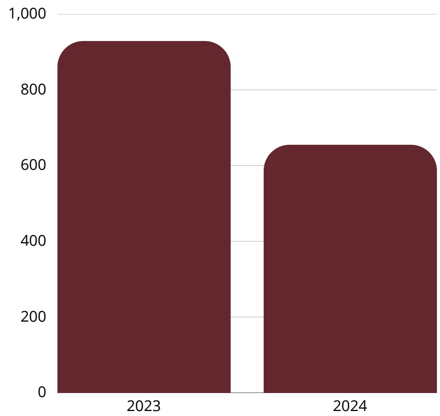
In practice, this means that founders often need to first build a basic understanding of the market among investors. Logistics and supply chains are fundamental to how the economy operates, yet they remain largely invisible to many people, so explaining their importance is often the first step in funding discussions.

At the same time, angel investors typically have limited capital and tend to spread their investments across multiple projects. In the case of venture capital funds, the issue is somewhat different. In the region, there is often a focus on quick returns rather than building scalable projects over a longer time horizon. Combined with limited sector knowledge, this results in transport and logistics projects, which require longer implementation cycles and deeper operational understanding, being less frequently considered for investment. Meanwhile, capital intensive sectors require large, stable, and patient sources of funding."

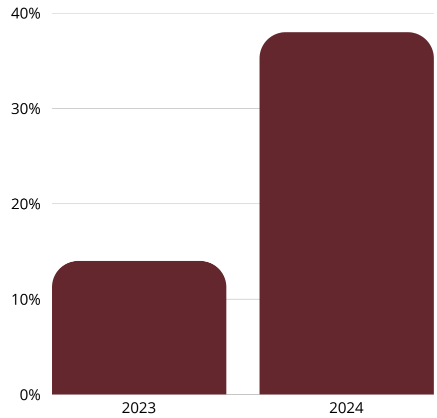
H1.3: The decline in mobility investments in Europe is driven by increasing investor caution and by the slowdown in the development of alternative propulsion technologies

Status: Confirmed | Confidence: High

In 2024, the number of VC transactions in European mobility declined from 929 to 655, a drop of 29%, with early stage transactions decreasing by around 30%. Capital shifted toward SaaS models, which accounted for 38% of investments compared to 14% the year before, while B2B models in mobility attracted 85% of total funding. In CEE, between 2020 and 2024, breakout and late stage rounds accounted for only 4% of transactions compared to 7% in Europe, highlighting a scaling gap.



Number of venture capital transactions in the mobility sector in Europe The data covers venture capital transactions in the European mobility sector based on the report “State of European Mobility Startups 2024”.
Source: Via ID, Dealroom



Share of the value of venture capital investments allocated to SaaS models in the mobility sector in Europe, %
The data covers venture capital investments in the mobility sector in Europe based on the report “State of European Mobility Startups 2024”.
Source: Via ID, Dealroom

Mathias Bosse’s comment:

„Investors have gone through a learning phase. Many logistics models were valued as software, while in reality they remain low margin businesses. Digitalization alone does not change the margin structure. As a result, funds are now much more cautious and conduct deeper analysis to assess whether a given model truly has venture scale potential”

Michał Lasocki's comment:

„The 30% decline in mobility funding in Europe in 2024 and the simultaneous increase in SaaS share from 14% to 38% are two separate phenomena. From a VC perspective, however, they reflect the same investment decision. Funds have not withdrawn from the sector, they have repriced risk.

Capital intensive models such as fleets, charging infrastructure, and hardware have been valued for what they are: low margin businesses with long return horizons and regulatory risk. Capital has shifted toward software that manages this infrastructure such as TMS, WMS, orchestration platforms, and route optimization, where the risk profile is closer to traditional B2B SaaS with low initial CAPEX, recurring revenue, and a faster path to unit profitability.

This distinction is important for the report, as it implies that mobility as an investment category has not shrunk, it has shifted. CEE has a real opportunity in this segment, given its strong engineering capabilities and lower product development costs compared to Western markets."

H1.4: Investments in MobilityTech are expected to increase significantly between 2025 and 2035

Status: Partially confirmed | Confidence: Medium

The transformation of mobility in Europe is strongly driven by EU regulations and public funding, which enforce fleet modernization and infrastructure expansion, generating structural investment demand further reinforced by urbanization and the development of autonomous transport.

At the same time, the pace of change may be constrained by geopolitical uncertainty, the risk of regulatory adjustments, and low margins combined with market fragmentation in CEE, which make the implementation of capital intensive projects more difficult.

H1.5. The lack of specialization among VC funds in the MobilityTech sector constitutes one of the main barriers to investment in the CEE region

Status: Partially confirmed | Confidence: Medium

Dealroom data shows that between 2019 and 2024, only 7 Series A+ rounds in mobility were completed in Poland and 49 across the entire CEE region, while in countries such as Germany or France this number reaches 30 to 40 annually. Over 70% of funds in CEE focus on Pre Seed and Seed stages, while only around 25% participate in Series A, creating a funding gap above EUR 10 million.

Maciej Stopczyk's comment:
„The main barrier to investment in Poland and more broadly in the CEE region is the near absence of specialized VC funds. Funds in the region lack sector specific expertise and patience, their investment horizons are significantly shorter than in mature ecosystems, and expectations for quick results are much higher. Meanwhile, capital intensive sectors require large, stable, and patient sources of funding. As a result, many logistics companies in Poland operate on the edge, relying on individual grants and hoping that commercial contracts will materialize in time to keep their projects alive."

Mathias Bosse's comment:
„Fund specialization emerges with market maturity. In Western Europe, there are funds with very clear theses focused on mobility or logistics. In CEE, this is still a work in progress. It is not a lack of potential, but rather a stage in the development of the ecosystem."

Sebastian Wróbel's comment:

„Ten years ago, conversations with investors often took place in a vacuum, neither founders nor funds had a shared language for mobility projects. Today, the situation is healthier: funds think more long term, understand scaling, and do not focus solely on quick equity stakes. This shows that the market is maturing. The issue is not the mindset of entrepreneurs or users, but the fact that the region's financial infrastructure developed later and faster than its experience."

Matthias Schanze's comment:

„Corporate capital can be great because it can be a proof point for startups that a corporate customer fully believes in them and offer a partnership to scale. It can also be a danger when the startup becomes dependent on one corporate. We try to address that challenge by being a VC fund that combines the speed and independence of a financial investor with the domain know-how and support of a strategic VC. This is also something that not only the startups appreciate, but also our corporate LPs."

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H2.1: Poland serves as one of the key transport and logistics hubs in Europe, but this does not translate proportionally into the development of innovation in the MobilityTech sector

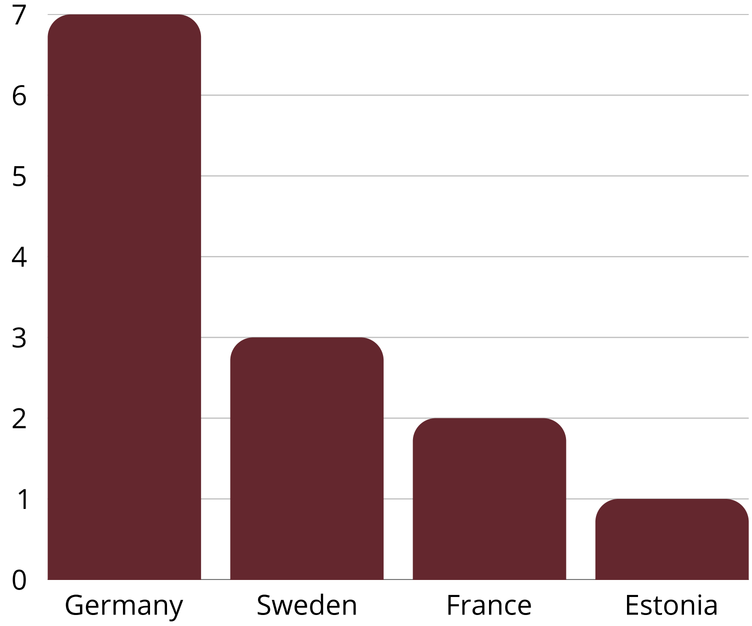
Status: Partially confirmed | Confidence: Medium

Poland is the EU leader in road freight transport, reaching 368 billion tonne kilometers in 2024, representing around 20% of total EU volume, compared to 281 billion in Germany and 272 billion in Spain. Despite this operational scale, none of the 15 European mobility unicorns originate from Poland, while Germany has 7, Sweden 3, France 2, and within CEE only Estonia is represented by Bolt.

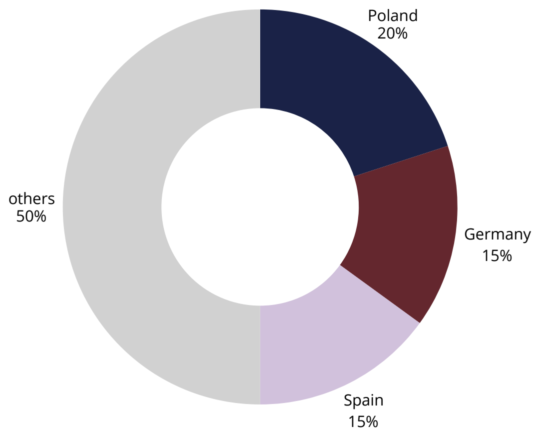
In the Polish technology ecosystem, there is a limited number of unicorn scale companies, among which InPost stands out as an example of a logistics company with international reach, originating from an early stage development model similar to a startup. InPost is typically classified as a mature public logistics company rather than a mobility startup, which is the main reason it is not included in the analyzed dataset.

Between 2019 and 2024, CEE attracted EUR 4 billion out of EUR 44.68 billion in VC investment in transportation in the EU, representing 9%, with more than half allocated to Estonia. This highlights a gap between Poland's operational strength in transport and the scale of innovation and mobility technology funding, although part of innovation takes place outside the VC market or through company relocation abroad.

Number of companies classified as “unicorns” in the mobility sector in Europe by country
Data based on Rethink Ventures analysis identifying companies with valuations exceeding €1 billion



Share of countries in the volume of freight transport measured in tonne kilometres in the European Union, %
Data for 2024. Source: Eurostat



Pavel Vrba’s comment:

„Poland and the broader CEE region do not suffer from a lack of talent, ideas, or capable teams. What is truly missing is risk capital that would allow these resources to translate into real innovation. This is particularly visible at later stages of growth. At the early stage, the situation is still relatively strong, but when a company needs substantial capital for expansion, the real challenge begins.

However, one of the biggest deficits in the region today is not only capital, but confidence. Solutions developed in CEE work, generate revenue, and have real market value, yet they are still too often presented with less confidence than comparable projects in Western markets. There is also a persistent fear of failure, whereas an unsuccessful first venture should not be seen as the end, but often as the beginning of the journey.”

Daniel Kierdal’s comment:

„Large international groups have R&D budgets, but key decisions are made at headquarters. Projects implemented locally typically need to demonstrate returns within a very short timeframe, often under two years. This limits the scope for more ambitious innovation.”

Sebastian Wróbel’s comment:

„In Poland, the TSL sector is highly fragmented. It consists mainly of small and medium sized enterprises that do not have the resources for R&D sufficient to cover even their own needs, let alone invest in external solutions.”

Tomasz Zarzycki’s comment:

„As Central and Eastern European countries, we are the operational backbone of European logistics, while still too rarely the owners of the technological value this sector generates. Our geographic position, scale of operations, and trends such as nearshoring and the shortening of supply chains give the region a real opportunity to become Europe’s logistics hub.

However, this requires a conscious shift away from ad hoc, “showcase” innovations toward solutions that deliver measurable business impact and fast returns. With margins at the level of 2 to 5%, automation and AI are no longer experiments but a competitive necessity. If we fail to seize this opportunity now, we risk remaining merely executors of others’ technological models rather than co creators of value in European logistics.”

H2.2: Structural barriers, such as low margins, market fragmentation, and capital intensity, limit the development of MobilityTech in the CEE region

Status: Confirmed | Confidence: High

The TSL sector in Poland operates on net margins of 2 to 5%, and nearly 70% of companies have fewer than 50 employees, including as many as 41% with no more than 10 employees. High capital expenditure limits investment, even though digitalization can reduce costs.

Adam Żelazko's comment:

"There are a large number of small players in the industry and very low margins. As companies scale, operational burden increases and absorbs those margins. Transformation and innovation are not only about creating new solutions, but above all about implementing them effectively. It is this stage that generates the highest cost.

When I speak with startups recommended by VC funds for validation, in our industry (TSL) I have practically not encountered anyone from the CEE region. Many technology companies scale globally, but they did not originate here. This is not a matter of lacking talented people, but rather that they either do not break through or there has not been an ecosystem and programs that would allow them to grow."

Pavel Vrba's comment:

„Poland's historical competitive advantage, based on low operational costs, is now becoming a constraint. Low margins translate into a reduced ability to reinvest in innovation and build a strong technology ecosystem. This challenge is visible not only in Poland, but across the broader Central and Eastern European region."

H2.3: Low awareness of innovation processes limits the effectiveness of investments

Status: Confirmed | Confidence: Medium

In the TSL sector in CEE, innovation is often ad hoc, and companies rarely run structured technology pilots, especially within the SME segment.

In Poland, there are around 20 accelerators, but the vast majority have a generalist profile. The gap between the needs of smaller companies and available technologies is further widened by the lack of mature innovation processes.

This gap is being addressed by selected industry initiatives such as StartSmart CEE, LogTech Innovation Platform, and MobiScale Incubator.

Adam Żelazko's comment:

"In the SME segment, day to day operational challenges dominate. Owners and managers do not have the time to pursue systematic improvements, and decisions to implement new tools typically emerge only at later stages of growth, when existing solutions are no longer sufficient."

Sebastian Wróbel's comment:
„Transport and logistics in CEE remain a fragmented and still relatively low digitized market. On one hand, this is a barrier, as implementations are difficult and costly, but on the other, it creates significant space for products that solve real problems across multiple stakeholders. Fragmented markets enable pivoting and product adaptation during development. For technology teams, this is not a disadvantage, but a strategic advantage, provided they can leverage it effectively."

Mathias Bosse's comment:
„In large logistics companies, innovation tends to be cyclical. Digital and venture units are created, only to be shut down after a few years, especially during periods of slowdown. Building an advantage through venture building requires a 5 to 10 year horizon, and few organizations are willing to commit to that level of patience."

H2.4: MobilityTech in the CEE region remains three to five years behind e-commerce, but the adoption of technology in the sector will accelerate

Status: Confirmed | Confidence: Medium

The AI market in logistics is growing very rapidly. According to GMInsights data, the AI market in logistics and supply chains reached USD 20.1 billion in 2024, with forecasts indicating growth at a CAGR of 25.9% between 2025 and 2034. At the same time, 90 percent of small transport companies do not use AI, and only 12% of larger companies with more than 50 employees have implemented it. According to Eurostat data for 2025, in CEE countries such as Romania, Poland, Bulgaria, and Hungary, only 5% to 10% of companies use AI, compared to over 40% in the most advanced EU country in this segment, Denmark, highlighting a significant gap and an urgent need to accelerate adoption.

Sebastian Wróbel's comment:

„Mobility in Poland and more broadly in CEE still lags behind other industries in terms of digitalization. It is a highly fragmented, operational market built on relationships and margins of just a few percent. In such conditions, any technological change is a high risk decision. This does not mean a lack of awareness, but rather a lack of financial and operational capacity to implement solutions faster."

Łukasz Chyliński's comment:

„The most important factor is awareness of new technologies, as the market landscape in TSL and emerging technologies is changing. It is no longer possible to improve performance through conventional methods such as hiring more drivers, cutting costs, or expanding fleets, as these approaches have largely been exhausted. Companies and entrepreneurs are now turning toward new technologies and looking at AI with both optimism and a certain degree of caution."

Daniel Kierdal's comment:

„The Polish transport market is extremely fragmented. Many companies still operate using Excel or systems developed over a decade ago by local software houses that never had ambitions to scale. This is not a lack of technology, but a lack of standardization and pressure to modernize."

Marcel Nawalany's comment:

„I believe that the assumption that the transport and logistics sector is only 3 to 5 years behind e commerce in terms of technology is too optimistic. In my view, the gap is significantly larger. At the same time, I see strong potential for acceleration in the coming years, primarily driven by generational change. Companies founded in the 1990s by the first generation of entrepreneurs are gradually being taken over by a younger generation of managers who are much more open to technology and process digitalization.

However, the biggest challenge remains the level of understanding of new technologies within organizations. Many concepts such as blockchain or artificial intelligence are present in the industry, but they often do not translate into real implementations. Technologies can be highly valuable, but the key question is whether companies are able to translate them into concrete business applications.

In the case of AI, it will be particularly important to overcome employee concerns and demonstrate that its role is not to eliminate jobs, but to support people by automating repetitive tasks and increasing efficiency."

Michał Wolański's comment:

„Logistics is still a sector waiting for its Uber, Booking, or FlixBus moment. The bus industry, which was also a low margin business, was completely transformed by FlixBus. However, there is a risk that this sector may never see such a disruptive innovation. Poland is a leader in logistics in Europe, particularly in road transport, which is why it would be disadvantageous if the digital leader of this industry were to emerge in another country."

H3.1: Regulations, urbanization, demographic trends, and labor costs will be key factors shaping the development of MobilityTech in CEE countries.

Status: Partially confirmed | Confidence: High

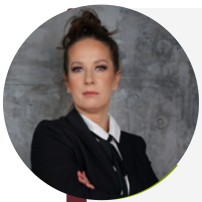
The EU aims to reduce emissions by 55% by 2030 and to ban the sale of new internal combustion engine cars from 2035. By 2050, more than 83% of EU residents will live in cities. In CEE countries such as Poland, the Czech Republic, and Hungary, wages have been growing at around 12% annually since 2022, increasing pressure for automation. However, this process may be slowed by regulatory uncertainty and uneven adoption rates across EU countries.

Adam Żelazko's comment:
„Looking at the demographics of Central and Eastern Europe through to 2050, companies will need to adopt technology to improve productivity. Even if they have not done so so far, they will be forced to, because scaling through additional workforce will become increasingly difficult to sustain. Companies must seek automation, as costs will rise while the availability of workers will decline."

Mathias Bosse's comment:
„Models driven solely by regulatory pressure, particularly in the area of sustainability reporting, have seen a clear decline in interest. When regulations were softened, some investments slowed down. This shows that sustainable business models must be built on operational value, not purely on compliance."

Łukasz Chyliński's comment:

„Twenty years ago, Polish transport companies built their competitive advantage in the European market thanks to access to drivers, at a time when this resource was already scarce in Western Europe. As shortages increased, the market was supplemented with workers from Ukraine and Belarus, and more recently also from Africa. Today, the situation has reversed. Companies have contracts and equipment, but are increasingly lacking drivers to operate trucks, which is pushing the industry to seek new ways to improve profitability."



Joanna Porath's comment:

„The reform of the EU customs system fits into the broader impact of regulation on the development of MobilityTech in CEE. Beyond emission reduction requirements and the energy transition, there is a growing importance of regulations related to data standardization and integration across the entire Union.

The planned shift toward more integrated systems, such as the EU Customs Data Hub, represents a move away from fragmented, local processes toward a model based on data management and data quality. The new model assumes the integration of trade, customs, and regulatory data within a single environment, supported by analytics and automation, effectively shifting the center of gravity from physical operations to data management.

In practice, this strengthens the pressure for automation, interoperability, and the implementation of data driven solutions. This direction aligns with the growing role of SaaS, automation, and process digitalization as the most rational investment path in a low margin environment.

As a result, regulations begin to play a dual role. On one hand, they generate transformation pressure, and on the other, they define the target operating model of the market, based on system integration and continuous data flow. In such a system, the key capability becomes the ability to ensure data consistency and high quality, as well as integration with both EU level systems and those of business partners."

BEDNARZ VENTURES

H3.2: Green mobility will be a significant catalyst for the transformation of the MobilityTech sector in the CEE region

Status: Partially confirmed | Confidence: High

Central and Eastern Europe as a potential hub for EV component manufacturing due to lower labor costs and a strategic location. CEE countries maintain a cost advantage, with productivity reaching around 80% of the EU average, while labor costs in most countries in the region are around 40–60% of the EU average, and in some countries, such as Romania, even around 35%, but the region still lags behind in infrastructure.

Paweł Michalski's comment:
"I do not see a clear breakthrough in green mobility in the region today. I believe there will be more investment, but rather driven by overall capital growth than by a deep structural shift. Projects such as IZERA show that despite strong ambitions, they have not generated a broader development impulse for the ecosystem.

In electromobility, strategic decisions have been made not to invest, which limits market momentum. Additionally, energy storage in our region remains several years behind Western Europe. Without real test beds and early stage customers, it is difficult to view green mobility as a catalyst for transformation. More likely, we will follow the change rather than lead it."

H3.3: Process automation will be driven by market consolidation and rising labor costs

Status: Confirmed | Confidence: High

In CEE, labor costs in 2023 increased by around 11% annually, while in 2023 to 2024 productivity in the region grew on average by only about 0.7%, weakening the competitiveness of the TSL sector. The logistics automation market was worth 17 billion USD in 2023 and may reach 49 billion USD by 2033, while implementation leaders increase efficiency by 20% to 40% within 2 to 4 years. The integration of digital tools can reduce EBITDA risk by 60% and increase valuation by 20%, however smaller companies still face limited access to capital and know how.

Adam Żelazko's comment:
"In the coming years, process automation in TSL will no longer be an option but a necessity. Rising technology costs and market consolidation will force investments in solutions that increase productivity, as this will be the only way for companies to maintain competitiveness and improve low margins."

Sebastian Wróbel's comment:
„I do not believe that user awareness in TSL is lower than in other industries. The difference lies in market structure and margin levels. Once cost pressure begins to materially impact profitability, technology adoption will accelerate faster than we currently expect."

Aleksandra Kaszub's comment:
"In operating companies, most resources are focused on day to day operations, while innovation remains an additional activity. At Marathon, we treat pilots as a management tool. They are embedded in our processes and have clearly defined business objectives. The key change for the market is the shift from a project based approach to continuous testing and scaling of solutions based on data"

Daniel Kierdal’s comment:
„The market is no longer growing as dynamically as before, and margins are steadily declining. Pressure on operational efficiency will naturally drive integration, APIs, and process automation. This will not be a technological trend, but a matter of survival.”

H3.4 MobilityTech is a strategic focus area of public policy in the context of the energy and climate transition

Status: Partially confirmed | Confidence: High

Transport accounts for approximately 25% of greenhouse gas emissions in the EU, of which around 73% is generated by road transport, while in Poland emissions from this sector increased by about 95% between 2005 and 2023. The EU requires a 90% reduction in transport emissions by 2050, while 90% of energy used in road transport in 2023 still came from fossil fuels.

Share of road transport in greenhouse gas emissions within the transport sector in the European Union, %
Data for 2023 based on the European Environment Agency.

Sebastian Pruckmair’s comment:
“The key question from an investor’s perspective is what regulations will require over the next 10 to 20 years. It is precisely these regulations and emission reduction trends that are shaping the direction of investment in mobility today.”

H4.1: AI, IoT, and 5G serve as the foundational technologies driving the development of the MobilityTech sector

Status: Confirmed | Confidence: High

AI is becoming a key tool for transforming transport and logistics, enabling route optimization, reduction of empty runs, and better fleet utilization without high CAPEX. According to the World Economic Forum, the combined application of operational efficiency improvements, capacity utilization optimization, and modal optimization can reduce emissions in freight transport by 10 to 15%. Barriers include high implementation costs for SMEs and the lack of unified data exchange standards, which hinders system integration and the scaling of AI solutions across the supply chain.

Aleksandra Kaszub's comment:
"The potential of AI solutions is real and achievable, which we see every day. Route optimization, reducing empty miles, and better fleet utilization are all concrete, measurable outcomes. The biggest challenge is not the technology itself, but scaling it through systems integration, data quality, and process consistency across the entire organization. Implementing a single use case is relatively straightforward, but scaling it across the whole organization requires operational maturity and a consistent approach to building a digital environment."

Kacper Nowicki's comment:
"Physical AI is an area that will bring fundamental changes over the next 10 years, not only in mobility but beyond. AI is increasingly entering the physical world, directly impacting real world operations."

H4.2 MaaS solutions benefit from urbanization and from changing societal expectations

Status: Confirmed | Confidence: Medium

Urbanization in CEE is among the fastest in the European Union, with population growth concentrated primarily in the largest metropolitan areas, which are expanding significantly faster than national averages. This trend is strengthening demand for integrated and flexible mobility models.

In 2024, shared mobility in Europe reached 640 million rides, representing a 5% year over year increase, with Poland ranking second in Europe in terms of the number of shared vehicles. The development of MaaS and the integration of public transport with micromobility are becoming a permanent element of urban strategies, although regulatory barriers and lack of scale continue to limit profitability in smaller cities.

Mathias Bosse's comment:
„Some mobility investments by VC were driven by hype cycles, such as digital freight forwarding or micromobility. In CEE, this effect was less pronounced, as the ecosystem was younger and less visible."

H4.3: The focus of investments in MobilityTech is shifting from EV hardware toward software and system interoperability

Status: Confirmed | Confidence: High

In 2024, the structure of mobility funding in Europe shifted significantly. Debt financing in capital intensive hardware segments reached USD 9 billion, while the share of SaaS models increased to 38% of total investment, up from 14% the year before. This indicates a shift toward scalable, asset light solutions based on software.

APIs play a key role by enabling system integration and real time process automation. However, the main barrier remains the lack of interoperability and limited willingness of companies to share data.

Jan Radzikowski's comment:
„The key factor is the geopolitical and macroeconomic environment, including global trade tensions and tariff disputes, rising risks of armed conflicts, and potential corrections in capital markets, where many assets are currently at historically high valuations.

In such conditions, investments with long return horizons and high capital intensity are the most exposed, as they require significant upfront capital under elevated systemic risk and uncertainty about the future economic environment."

Daniel Kierdal's comment:
„Many solutions are developed as projects for a single large client. The most difficult moment is turning such a project into a scalable market product. When the roadmap is dominated by the needs of one customer, it is easy to lose universality and scalability potential."

H5.1: Public and corporate capital dominate investments in the MobilityTech sector

Status: Confirmed | Confidence: High

n Poland and across CEE, corporate VC funds play a leading role in financing mobility. Orlen VC has announced a budget of EUR 100 million over a 10 year horizon, while also running a non-equity accelerator program. PGE Ventures, through the SpeedUp Energy Innovation fund with a budget of PLN 60 million, invests in areas such as electromobility, smart cities, and industrial automation.

The activity of private mobility focused VC funds remains limited due to capital intensity, long implementation cycles, and regulatory risk, which drives capital concentration toward SaaS models. At the same time, in 2023, 92% of 818 VC transactions in CEE were at the pre seed and seed stages.

Mathias Bosse's comment:

"From the perspective of the German market, MobilityTech clearly shows a dominance of later stage rounds and a strong presence of corporate VC. Corporate funds are not comfortable investing at very early stages and prefer to enter when the business model is more mature and the strategic component becomes relevant.

As for the CEE region, I primarily see an issue of visibility and lower ecosystem maturity, rather than lower founder quality. Dealflow from the region is smaller, there is limited presence of Western European VC, and the ecosystem infrastructure is less developed than in Germany and across Western Europe."

Sebastian Wróbel's comment:

"In Poland, VC funds did not mature in the same way as in Western Europe. One of the mechanisms supporting market development has been the introduction of co investment structures and capital leverage by public investment vehicles, which partially reduced risk on the investor side. As a result, part of private capital has concentrated in funds benefiting from such mechanisms.

This may influence market structure and lead to a situation where some very early stage companies have more limited access to external funding, as investors may find it more attractive to allocate capital within structures where part of the risk is shared with public capital."

H5.2: There is a gap in the CEE region for infrastructure focused deeptech mobility funds

Status: Confirmed | Confidence: High

In CEE, there is a lack of specialized mobility deeptech funds, and most of the nearly 200 VC funds analyzed operate under a generalist model without sector specific mandates, based on the 11VC report. According to data published by Dealroom, startups founded in the CEE region often relocate as they scale. The data shows that nearly half, 48%, of CEE scaleups have established their headquarters outside their country of origin. Among companies that chose to relocate, 56% moved to the United States, while 39% relocated to other European countries, including 24% to the United Kingdom.

Kacper Nowicki's comment:

"If a founder from the CEE region decides to build a global company, they need to consciously choose where to raise capital. If they raise in Warsaw, they will typically receive a lower valuation for the exact same project than in Berlin, London, or the United States. This is not a matter of talent or product quality, but of capital market scale and network.

Access to events in Berlin, Paris, or London is now effectively the same as access to those in Poland. The difference lies in whether the founder chooses to be present there, build relationships, and optimize for larger pools of capital, or opts for convenience. If someone has a strong value proposition and a solid pitch, they can raise capital outside their home country. The key is being willing to go where that capital is."

Daniel Kierdal's comment

„Most Polish funds operate at the seed stage. As companies progress to later rounds, they need to align with the expectations of international investors. This encourages copying trends from Western markets rather than building specialized sector expertise."

Matthias Schanze's comment

„We are a Mobility & Logistics focused VC and we are happy to invest in deep tech and infrastructure. We do not see that there is a lack in funding, but in deed a lack of financial VCs that are able and willing to look deeper into deep tech, particularly in complex industries such as logistics or transportation. Concerning infrastructure: VCs are increasingly putting this field on top of their investment agenda, yet it still needs to pose a scalable business model and VC-like investment case."

H5.3: The sector is undergoing a resilience test driven by AI, which results in slower scaling in the B2B enterprise segment but leads to more stable revenues

Status: Partially confirmed | Confidence: Medium

AI adoption in Polish logistics remains at a very early stage despite high awareness of its benefits. 57% of companies expect process acceleration and 40% expect cost reduction, yet only around 21% actually use AI. The main barriers include implementation costs at 55%, integration challenges at 39%, and concerns about the pace of technological development at 44%. According to GUS, 94.1% of Polish companies did not use AI in 2024, while in logistics only 0.6% of enterprises declared its use. Across CEE, AI adoption is around 10% lower than the global average, with a skills gap reaching approximately 25% below the global benchmark.

Grzegorz Stańczak's comment:

„I see a certain duality in the industry. On one hand, there is awareness of AI's potential, on the other, a high level of caution. There is still lack of understanding, reluctance, and a perception of relatively high risk, which I personally do not share. I also feel that we lack tangible examples of returns from long term investments. It seems that few have seen a real long term R&D project in this space that delivered clear value and returns. Despite this, I believe there is no turning back. The technology will not stop evolving, and it is better to start adopting, exploring, understanding, implementing, and above all using it, rather than blocking it out of fear."

Michał Lasocki's comment:

„Data suggesting that 94% of Polish companies do not use AI is technically correct, but methodologically flawed. GUS surveys and similar studies ask directly whether companies use AI, and firms often respond no because they do not identify the tools they use as AI. In reality, a significant share of AI implementation in logistics and transport is embedded within TMS and WMS platforms, including load optimization, predictive maintenance, and dynamic routing, without any explicit AI project on the client side. Companies purchase a fleet management system and effectively receive AI as part of the package. From our observations in the energy sector, we see the same effect. Actual adoption rates measured by system providers are several times higher than those declared in surveys. This suggests that the adoption gap presented in reports may be overstated, which is important as it affects the perceived market potential for new AI native entrants. This does not change the fact that AI remains a domain with one of the highest failure rates in IT projects, largely due to still limited practical understanding of the technology."

Natalia Nogacka's comment:

„In logistics, the problem today is not the capabilities of AI tools themselves, but the way they are implemented and embedded within organizations. Despite growing awareness of the potential of automation, faster data processing, and support for operational decision making, companies often enter this area without structured processes, with fragmented systems, and without clearly defined responsibility for implementation.

Therefore, in the coming years, it will not be the organizations that declaratively express interest in AI that succeed, but those that are able to translate it into concrete, integrated operational processes, embedded in everyday work and held accountable for measurable business outcomes."

Aleksandra Kaszub's comment:

"When it comes to implementing AI based solutions in the TSL industry, we are facing a paradox. AI is already at work in many organizations, but it is not referred to as AI. Systems supporting planning, forecasting, or route optimization often incorporate AI capabilities, yet they operate as part of broader software platforms. The real difference lies in whether a company consciously builds its competitive advantage around data and automation, or simply relies on the built in features of existing systems.

Today, the main barrier to AI implementation is not a lack of awareness, but organizational readiness. Many companies attempt to deploy AI on top of fragmented data and poorly structured processes, which often leads to disappointing results. The real turning point comes when leadership sees a tangible impact on business performance, for example, improved margins through better planning optimization. At MARATHON, AI is not an experimental project. It is a natural extension of well structured operational processes."

Implications for decision makers in the MobilityTech sector

CEE has a strong operational position in European logistics and transport, yet remains on the defensive when it comes to capturing technological value.

This tension will intensify in the coming years, driven by three simultaneous forces: rising labor costs, regulatory pressure, and technological pressure. In this context, innovation in TSL companies is no longer a question of whether, but how: how to invest in technology in an environment of low margins, fragmentation, and limited CAPEX, and how to do so without falling into the trap of startup theatre, meaning a series of costly, non repeatable experiments without scale.

The most rational entry point is where returns are measurable and implementation can be standardized: automation and AI in back office functions and operational micro processes. These areas are typically underinvested, yet sufficiently repeatable to enable scalability.

The report assumes that back office and cash processing may account for a few percent of revenues, while underinvested process segments represent a significant share of operator costs. With net margins at 2 to 5%, investments that do not generate measurable impact within a 12 to 24 month horizon are systematically deprioritized. Therefore, the entry point should be pragmatic: focus on micro processes that can be structured, measured, and gradually automated, rather than attempting full organizational transformation from the outset.

The second implication concerns risk, and in practice it will determine the pace of adoption. In a world where AI is accelerating and integrations, data, and cybersecurity are becoming increasingly complex, technology decisions must be evaluated not only through the lens of budget and ROI, but above all through implementation risk. Our analysis shows that integration costs and complexity are among the main barriers to AI adoption.

If the market is to accelerate, it requires implementation models that reduce the cost of failure: standardized pilots, clear success criteria, shorter decision cycles, and risk sharing mechanisms between technology providers and adopters. This creates a natural role for innovation consortia, partnerships, and PPP type projects, not as innovation marketing, but as tools for risk reduction in a sector with low tolerance for unsuccessful CAPEX.

The third implication is institutional and relates to scaling. In the context of a funding gap at the Series A+ level and limited fund specialization, startups alone will not fill the region's technological deficit.

If CEE aims to build its own projects at a European scale, it needs market infrastructure that removes key bottlenecks: access to pilot environments, data interoperability, integration capabilities, and enterprise deployment channels. Without this, the most promising teams will continue to relocate as they scale, and the region will remain more of an executor than an owner of value. In the long term, this implies a widening competitive gap with Western markets, especially under rising labor cost pressure.

Therefore, the key is to think in terms of systems, not just capital, systems that enable the transition from one off implementations to repeatable and scalable products, similar to how certain sectors in Poland have built shared market standards, such as BLIK as a model of coordination and large scale adoption.

Playbook 2025-2035.

Three scenarios for the development of the MobilityTech sector in CEE

If the report's diagnosis is accurate, and both data and market conversations suggest that it is, then the future of MobilityTech in CEE will not be driven by a single breakthrough project or one major funding round. It will depend on whether the region can simultaneously activate several systemic levers like capital, implementation standards, industry collaboration, and scaling infrastructure.

The scenarios below are not forecasts. They represent possible trajectories directly derived from current constraints: CEE's low share in MobilityTech VC value, the gap in follow on funding, low AI adoption, net margins of 2 to 5% in TSL, and market fragmentation.

Scenario 1

Status quo maintaining cost advantages without building technological advantages

In this scenario, the market structure remains largely unchanged. TSL companies continue to compete based on operational efficiency, geographic location, and labor costs, while technology investments are implemented mainly as isolated projects or through global system providers, often funded by EU programs as internal initiatives of market players.

VC funds remain largely generalist, focused on pre seed and seed stages, and the Series A+ gap persists. Projects with international ambitions continue to relocate as they scale, as confirmed by Dealroom data showing that 48% of CEE scaleups move their headquarters outside the region.

The consequence of this path is a further widening gap between operational strength and the ability to capture technological value. CEE remains an executor of logistics processes for Europe but does not build its own platforms or venture scale products. In an environment of rising labor costs around 11 to 12% annually and increasing regulatory pressure, this results in sustained margin pressure without technological compensation.

Scenario 2

Moderate transformation selective adoption and gradual ecosystem maturation

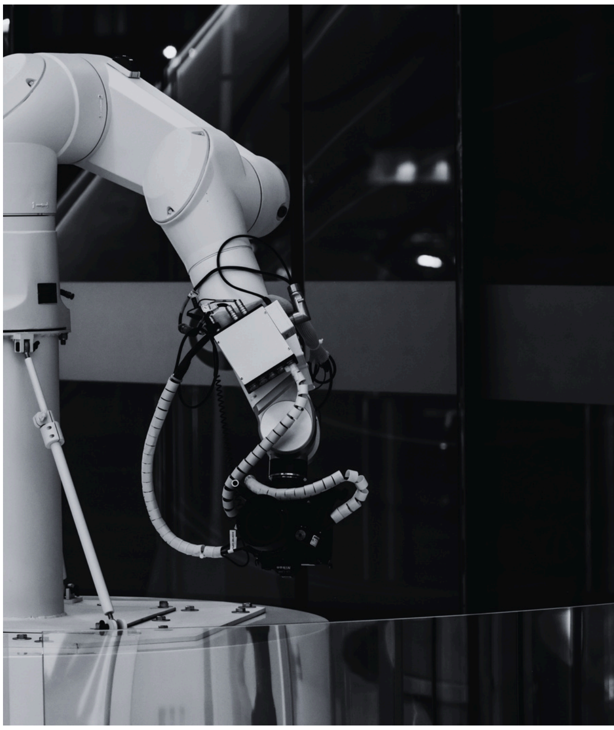
The second scenario assumes a pragmatic approach aligned with the implications outlined in the report. Transformation begins in areas with measurable ROI and potential for standardization: automation of micro processes, back office digitalization, and AI deployment in clearly defined operational use cases.

In this scenario:

- some funds develop specialization in mobility and climate,
- operator and startup consortia emerge around shared pilot projects,
- the role of corporate VC grows as a stabilizing source of funding,
- integration standards and data interoperability become part of the industry agenda.

CEE gradually converges toward the EU average in MobilityTech VC share at around 8 to 9%, AI adoption among large TSL companies increases to 40 to 60%, and one or two unicorn scale projects emerge in the region.

This is a realistic and achievable path without the need for institutional disruption. However, it requires consistency, a long term approach to venture building, and a shift away from single client driven projects toward scalable product models.



Scenario 3

Coordinated systemic shift from fragmented initiatives to an integrated ecosystem

The third scenario represents a shift from a project based logic to a system based approach. It requires the simultaneous activation of three levers:

1. Capital establishment of specialized mobility funds including growth stage, increased capitalization of CEE funds, and integration of corporate VC with public programs
2. Implementation infrastructure standardized pilot models, access to testing environments, data interoperability, and integration support
3. Industry coordination sector focused incubators and collaboration platforms connecting operators, technology providers, and regulators, similar to models in more mature ecosystems such as France

In this scenario, CEE's share in MobilityTech VC value could increase to 12 to 15%, AI adoption among large companies could exceed 70 to 80% within a few years, and the region would begin to build its own infrastructure and platform products rather than primarily adapting external solutions.

This scenario requires the highest level of institutional maturity and decision making courage. It implies a shift away from short budget horizons of 12 to 24 months toward a long term perspective of 5 to 10 years.

Levers that will determine the future

From a strategic perspective, three key areas emerge directly from the previous diagnosis:

- **Public policy and regulation** not as a source of hype, but as a mechanism for creating stable investment frameworks such as recovery funds, grants, PPP, and tax incentives. Regulation should reduce implementation risk rather than merely generate reporting obligations
- **Access to capital** greater VC specialization, development of growth stage funds supported by an increasing number of exits, expansion of industry focused corporate VC, and debt or grant instruments for automation projects. Without this, the growth stage funding gap will persist and relocation will remain a natural outcome
- **Partnerships and coordination** consortia of operators and technology providers, shared pilot standards, risk sharing mechanisms, and the development of interoperability infrastructure. In an environment of 2 to 5% margins and limited CAPEX, a single company cannot bear the full cost of experimentation on its own.

Ultimately, the choice of trajectory is not a decision made by a single actor. It is the result of the region's collective ability to move from a model of fragmented entrepreneurship to one of coordinated value creation.

If CEE relies on only one lever, for example increasing capital supply without building implementation infrastructure, the impact will remain limited. However, if all three areas are activated in parallel, MobilityTech in CEE can evolve from being Europe's operational backbone to becoming a co creator of technological value.

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